

MCQ.

13.6 Directional Derivative.

unit vector $\hat{a}$ convert ∇f $\rightarrow$ (বের করতে হবে)

$$\vec{\nabla} f = \frac{\partial}{\partial x}(f) \hat{i} + \frac{\partial}{\partial y}(f) \hat{j}.$$

perpendicular of function = corresponding gradient.

normal vector = gradient.

$$\begin{aligned}\vec{\nabla} f &= \left\langle \frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, \frac{\partial f}{\partial z} \right\rangle \\ &= \langle f_x, f_y, f_z \rangle\end{aligned}$$

$$f_x(x-x_0) + f_y(y-y_0) + f_z(z-z_0) = 0$$

$\hookrightarrow$ Tangent line plane of the eqn ellipsoid.

$$P_0 = \langle x_0, y_0, z_0 \rangle.$$

$$\vec{q} = \langle a, b, c \rangle.$$

gradient of f = parallel vector.

$$x = x_0 + f_x t.$$

$$y = y_0 + f_y t.$$

$$z = z_0 + f_z t.$$

} normal line.

Example 1. a)

$$x^2 + 4y^2 + z^2 = 18.$$

$$f_x = 2x = 2 \times 1 = 2$$

$$f_y = 8y = 8 \times 2 = 16$$

$$f_z = 2z = 2 \times 1 = 2.$$

$$f_x(x-1) + f_y(y-2) + f_z(z-1) = 0.$$

$$2x(x-1) + 8y(y-2) + 2z(z-1) = 0.$$

$$\Rightarrow 2x(x-1) + 16(y-2) + 2(z-1) = 0$$

$$\Rightarrow 2x - 2 + 16y - 32 + 2z - 2 = 0.$$

$$\Rightarrow 2x + 16y + 2z - 36 = 0.$$

$$\Rightarrow x + 8y + z - 18 = 0.$$

b)

$$x = x_0 + f_x t$$

$$y = y_0 + f_y t$$

$$z = z_0 + f_z t$$

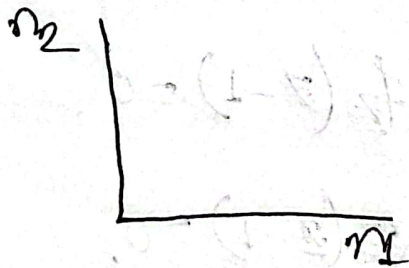
c) $\langle 2, 16, 2 \rangle$ and $\langle 0, 0, 1 \rangle$ → calculate acute angle between them.

Example 2.

find tangent plane $z = x^2 y$ at the point $\langle$

x, y, z

Two surface's normal vector.



$f(x, y, z)$ and $g(x, y, z)$ } one's intersecting point or tangent line

$$\nabla f(x, y, z) \times \nabla g(x, y, z) = 12i - 14j - 4k$$

Example 3

Linear Approximation: $f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$

$$f'(x) \approx \frac{f(x+h) - f(x)}{h}$$

$$h f'(x) \approx f(x+h) - f(x) \rightarrow \text{functional value.}$$

$$f(x) \approx f(x+h) - h f'(x)$$

$$f(x, y) = f(x_0, y_0) + f_x(x_0, y_0)(x - x_0) + f_y(x_0, y_0)(y - y_0)$$

Linear Approximation.

$$z = 3 + \frac{x^2}{16} + \frac{y^2}{9} ; (-4, 3)$$

$$z = 3 + \frac{x^2}{16} + \frac{y^2}{9}$$

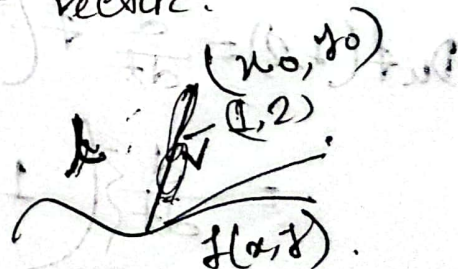
$$f(-4, 3) = 5$$

$$f_x(x, y) = \frac{2x}{16} = \frac{x}{8} = \frac{1}{2}$$

$$f_y(x, y) = \frac{2y}{9} = \frac{2}{3}$$

$u = u_1 i + u_2 j$ is a unit vector.

Directional Derivative



a line l in the xy -plane that can be expressed parametrically.

$$x = x_0 + u_1 s, \quad y = y_0 + u_2 s \quad \text{Let } \frac{d}{ds} (x_0 + u_1 s, y_0 + u_2 s)$$

$$f(x, y) = xy.$$

$$Du f(1, 2)$$

$$u = \frac{\sqrt{3}}{2} i + \frac{1}{2} j$$

$$\underline{u} = \left\langle \frac{\sqrt{3}}{2}, \frac{1}{2} \right\rangle.$$

$$(x_0, y_0) = (1, 2).$$

Def

$$x = x_0 + u_1 t.$$

$$x = 1 + \frac{\sqrt{3}}{2} t.$$

$$\boxed{x = 1 + \frac{\sqrt{3}}{2} t.}$$

$$\boxed{y = 2 + \frac{1}{2} t.}$$

$$Du f(1, 2) = \frac{d}{dt} f\left(1 + \frac{\sqrt{3}}{2} t, 2 + \frac{t}{2}\right).$$

$$= \frac{d}{dt} \left\{ \left(1 + \frac{\sqrt{3}}{2} t\right) \left(2 + \frac{t}{2}\right) \right\}$$

$$= \frac{d}{dt} \left(2 + \sqrt{3} t + \frac{t}{2} + \frac{\sqrt{3} t^2}{4} \right)$$

$$= \frac{d}{dt} \left(\frac{1}{2} \left(\sqrt{3} + \frac{1}{2} + \frac{\sqrt{3}}{2} t \right) \right) = \sqrt{3} + \frac{1}{2}.$$

$$D_u f(x_0, y_0, z_0) = \frac{d}{dt} [f(x_0 + u_1 t, y_0 + u_2 t, z_0 + u_3 t)]_{t=0}$$

$$D_u f(x_0, y_0) = f_x(x_0, y_0) u_1 + f_y(x_0, y_0) u_2$$

$$D_u f(x_0, y_0) = f_x(x_0, y_0) \cos \phi + f_y(x_0, y_0) \sin \phi$$

Example 2

$$f_x(x, y) = y e^{xy} \cdot \frac{d}{dx} (e^{xy})$$

$$f_x(x, y) = e^{xy} \frac{d}{dx} (xy) = y e^{xy} = 0$$

$$f_y(x, y) = x e^{xy} = -2 e^0 = -2$$

$$u = 1 \cos \frac{\pi}{3} i + 1 \sin \frac{\pi}{3} j = \frac{1}{2} i + \frac{\sqrt{3}}{2} j$$

$$D_u f(-2, 0) = \left(0 \times \frac{1}{2}\right) + (-2) \times \frac{\sqrt{3}}{2} = -\sqrt{3}$$

Example 3:

$$f(x, y, z) = x^2y - yz^3 + z.$$

$$(x_0, y_0, z_0) = (1, -2, 0)$$

$$a = 2i + j - 2k.$$

$$f_x(x, y, z) = 2xy = 2 \cdot 1 \cdot (-2) = -4.$$

$$f_y(x, y, z) = x^2 - z^3 = 1^2 - 0^3 = 1.$$

$$f_z(x, y, z) = 1.$$

$$a = 2i + j - 2k.$$

$$\|a\| = \sqrt{2^2 + 1^2 + (-2)^2} = \sqrt{9} = 3.$$

$$\frac{a}{\|a\|} = \frac{2}{3}i + \frac{1}{3}j - \frac{2}{3}k.$$

$$-4 \cdot \frac{2}{3} + 1 \cdot \frac{1}{3} - \frac{2}{3}$$

$$= -\frac{8}{3} + \frac{1}{3} - \frac{2}{3} = \frac{-8+1-2}{3}$$

$$= \frac{-9}{3} = -3$$

$$\vec{\nabla} = \frac{\partial}{\partial x} v \hat{i} + \frac{\partial}{\partial y} v \hat{j} + \frac{\partial}{\partial z} v \hat{k}$$

↪ gradient,

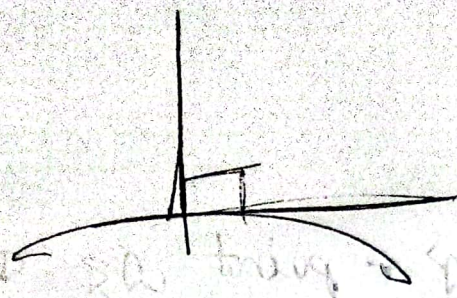
gradient 0 ରେ ∇ point ଏବଂ directional derivative ସର୍ବାଧିକ ହୁଏ ।

largest directional derivative = $\|\nabla f\|$ at P .

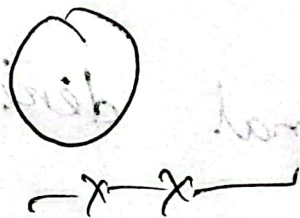
The smallest directional derivative = $-\|\nabla f\|$ at P .

Find the unit vector in the direction in which the maximum value occurs.

$$\frac{\nabla f(x_0, y_0, z_0)}{\|\nabla f(x_0, y_0, z_0)\|}$$



Example 6



13.7.

(a) $f(x, y, z) = x^2 + y^2 + z^2 - 18$

$(1, 2, 1)$.

$f_x(x-x_0) + f_y(y-y_0) + f_z(z-z_0) = 0$

$n = \langle f_x, f_y, f_z \rangle$

(b)

$$\left. \begin{aligned} x &= x_0 + f_x t \\ y &= y_0 + f_y t \\ z &= z_0 + f_z t \end{aligned} \right\}$$

Example 2

$$Z = x^2 y$$

$$f(x, y, z) = Z - x^2 y$$

$$f_x(x, y, z) = -2xy$$

$$f_y(x, y, z) = -x^2$$

$$f_z = 1$$